

GIZMO Operations Guide

Never lift the safety ground. Follow SOP004 and the current work plan.

Installed system

GIZMO is already deployed. SOP004 authorizes its installation and use. The initial grounding setup will use one saturable inductor.

Item	Recorded setting
Excitation	1436 Hz
Alarm threshold	100 Ω
Runtime / OPC UA model	0.5.1 / 1.4.0
Monitoring server	dune-fd-test01
History available	Since July 27, 2026

These are stored settings, not live confirmation. As of September 30, 2026, the instrument is unreachable from the monitoring server. The latest retained sample is September 13, 2026, at 07:11:34 UTC. The server dashboard and historian are running.

Connections after installation or maintenance

With the rack de-energized, check each cable against its installed label and the approved drawing:

- Pearson CT output to GIZMO CT input; the CT observes the intended safety-ground path.
- Stimulus center to the designated Detector Ground point; return to the designated rack/infrastructure reference.
- Beacon and audible output to their labeled connectors.
- Network to the assigned controls port; approved DC supply to the chassis.
- One saturable inductor in the initial setup, with its protection and safety-ground connections verified by the electrical lead.

Changes to grounding, CT routing, stimulus, or calibration require the applicable work controls. Do not improvise a bypass or move connections to clear an alarm.

Contractor grounding and isolation tests

The contractor will work with the cryostat and beam structure connected to Building Ground. GIZMO will check for shorts, capacitance, and coupling between Detector Ground and Building Ground during periodic isolation tests.

The planned arrangement uses a knife switch with tagout to establish the isolation-test state. The switching procedure must identify the switch, test position, protection, and responsible person. Do not treat an intentional construction bond as an isolation measurement.

1. Contractor work: verify and record the Building Ground connection while personnel are working on the structure.
2. Isolation test: coordinate a work pause, use the authorized knife-switch/tagout procedure, and record the actual grounding state and test time.
3. With fresh, usable GIZMO data, record the isolation indication, capacitance where valid, and coupling diagnostics. Compare with the permitted isolation limits.
4. Before contractor work resumes, restore and verify the required Building Ground connection, complete tagout closeout, and record the handback.

Schedule tests often enough to detect unwanted connections before further construction makes correction disruptive. Test cadence, shift routine, settling time, acceptance limits, and response to a failed or missed test: TBD. The recorded 100 Ω alarm threshold is not an approved construction-isolation acceptance limit.

Start of shift

1. Acknowledge the grounding state at shift start: contractor work with Building Ground connected, isolation test under tagout, or unresolved. Confirm the physical connection, knife-switch position, tagout status, and handover with the responsible lead.
2. Review the last isolation test, unresolved findings, and the next test due under the work plan.

3. Check the dashboard sample time, connection, quality, alarm, and recorded settings. Old data or a clear alarm alone cannot confirm isolation.
4. Record the acknowledgement and work state in Procore Daily Log > GIZMO. Detailed shift routine: TBD.

HIGH Z is a range indication, not a resistance value. The last stored sample had Uncertain quality, an uncertain clear alarm, and a latched alarm; it does not establish the present ground condition.

If an alarm appears

1. Check the recorded grounding state and any intentional construction or welding bond. An isolation result is not meaningful while the required Building Ground connection is in place.
2. If the alarm is unexpected for the recorded grounding state, stop new ground-affecting work. Record the time and alarm reason; notify the activity coordinator and grounding/GIZMO contact.
3. Check what changed: welding returns, temporary leads, cable shields, piping, scaffolding, or tools. Investigate under the work plan.
4. Keep the data and record the disposition. Do not change the threshold to make the alarm disappear.

If data stops

Treat stale or non-Good data as unavailable monitoring. A running dashboard or an old clear alarm is not proof that GIZMO is working.

1. Record the last sample time and notify the GIZMO and SC/DPS contacts.
2. During an isolation test, do not accept a result without valid data or an approved alternative. During contractor work, verify the required Building Ground connection; continuation and rescheduling follow the work plan.
3. Check the instrument power and network through the responsible on-site contact. The server reports loss of reachability; the cause has not been established.
4. After connection returns, verify advancing timestamps, usable measurement/alarm quality, and resumed history before relying on GIZMO again.

Welding

Use the approved welding-return arrangement. Record additional bonds and any approved alarm inhibition; keep raw alarms and history. Restore the contractor grounding state after work and check isolation at the scheduled test.

Server check

On dune-fd-test01, run:

```
/storage/gizmo-monitor/bin/status
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It reports the historian, dashboard, upstream connection, stored samples, and disk space. Check upstream connectivity and the newest GIZMO sample, not just whether the processes are running.

For a local dashboard connection:

```
ssh -N -L 127.0.0.1:18081:127.0.0.1:18081 dune-fd-test01
```

Then open <http://127.0.0.1:18081/> in the browser. Use the configured SSH access for the server.

Shutdown

Coordinate any loss of monitoring before shutdown. Use SOP004 and the deployed operating procedure to disable excitation and power down. Leave the permanent safety-ground path intact. Record the time and reason.